ECON484 project

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Predicting Employment Outcomes among Young Adults: Education, Well-being, and Machine Learning Evidence from PSID-TAS 2023.

```r
library(readxl)
```

```
## Warning: package 'readxl' was built under R version 4.5.2
```

```r
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.2
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

```r
library(glmnet)
```

```
## Warning: package 'glmnet' was built under R version 4.5.2
```

```
## Loading required package: Matrix
```

```
## Loaded glmnet 5.0
```

```r
library(randomForest)
```

```
## randomForest 4.7-1.2

## Type rfNews() to see new features/changes/bug fixes.

##
## Attaching package: 'randomForest'

## The following object is masked from 'package:ggplot2':
##
##     margin

## The following object is masked from 'package:dplyr':
##
##     combine
```

```r
library(rpart)
```

```
## Warning: package 'rpart' was built under R version 4.5.2
```

```r
library(rpart.plot)
```

```
## Warning: package 'rpart.plot' was built under R version 4.5.2
```

```r
library(gbm)
```

```
## Warning: package 'gbm' was built under R version 4.5.2

## Loaded gbm 2.2.3

## This version of gbm is no longer under development. Consider transitioning to gbm3, https://github.c
```

```r
library(nnet)
```

```r
data <- read_excel("/Users/natalieli/Desktop/ECON484 Project/2023/73datas.xlsx")

dim(data)
```

```
## [1] 2434   73
```

```r
names(data)
```

```
##  [1] "TAS"      "TAS23"    "ER30001"  "ER30002"  "ER32000"  "ER32006" 
##  [7] "ER35101"  "ER35102"  "ER35103"  "TA230001" "TA230050" "TA230051"
## [13] "TA230057" "TA230058" "TA230059" "TA230060" "TA230061" "TA230062"
## [19] "TA230063" "TA230065" "TA230066" "TA230067" "TA230069" "TA230070"
## [25] "TA230072" "TA230074" "TA230113" "TA230171" "TA230239" "TA230240"
## [31] "TA230241" "TA230244" "TA230427" "TA230428" "TA230628" "TA230629"
## [37] "TA230636" "TA230685" "TA230691" "TA230694" "TA230707" "TA230715"
## [43] "TA230716" "TA230861" "TA230948" "TA230950" "TA230953" "TA230954"
## [49] "TA230955" "TA230956" "TA230986" "TA230988" "TA230990" "TA230991"
## [55] "TA231000" "TA231001" "TA231025" "TA231145" "TA231146" "TA231147"
## [61] "TA231229" "TA231234" "TA232095" "TA232303" "TA232320" "TA232321"
## [67] "TA232322" "TA232325" "TA232335" "TA232392" "TA232393" "TA232394"
## [73] "TA232396"
```

```r
project_data <- data %>%
  mutate(
    employed = ifelse(
      TA230239 %in% c(1,2),
      1,
      0
    )
  )
table(project_data$employed)
```

```
## 
##    0    1 
## 1043 1391
```

Wellbeing

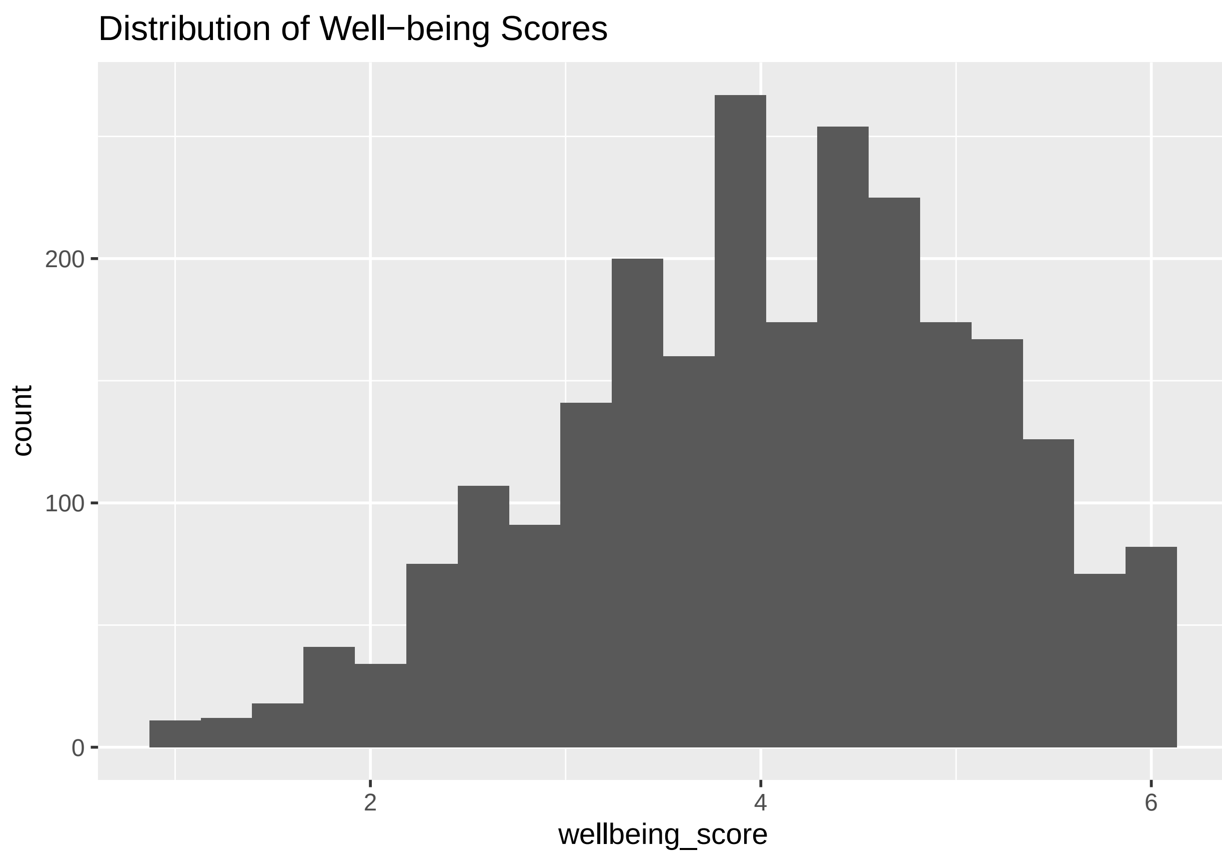
```r
#TA230061 happy
#TA230062 interest in life
#TA230063 satisfied
#TA230065 community
#TA230066 society better
#TA230067 people good
#TA230069 manage responsibilities
#TA230070 trusting relationships
#TA230072 confidence
#TA230074 life purpose

wellbeing_vars <- c(
  "TA230061",
  "TA230062",
  "TA230063",
  "TA230065",
  "TA230066",
  "TA230067",
  "TA230069",
  "TA230070",
  "TA230072",
```

```r
  "TA230074"
)

project_data <- project_data %>%
  mutate(
    across(
      all_of(wellbeing_vars),
      ~ifelse(. == 9, NA, .)
    )
  )

project_data <- project_data %>%
  rowwise() %>%
  mutate(
    wellbeing_score =
      mean(
        c_across(all_of(wellbeing_vars)),
        na.rm = TRUE
      )
  ) %>%
  ungroup()
summary(project_data$wellbeing_score)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##   1.000   3.400   4.100   4.062   4.900   6.000       4
```

```r
ggplot(
  project_data,
  aes(x = wellbeing_score)
) +
  geom_histogram(
    bins = 20
  ) +
  labs(
    title = "Distribution of Well-being Scores"
  )
```

```
## Warning: Removed 4 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



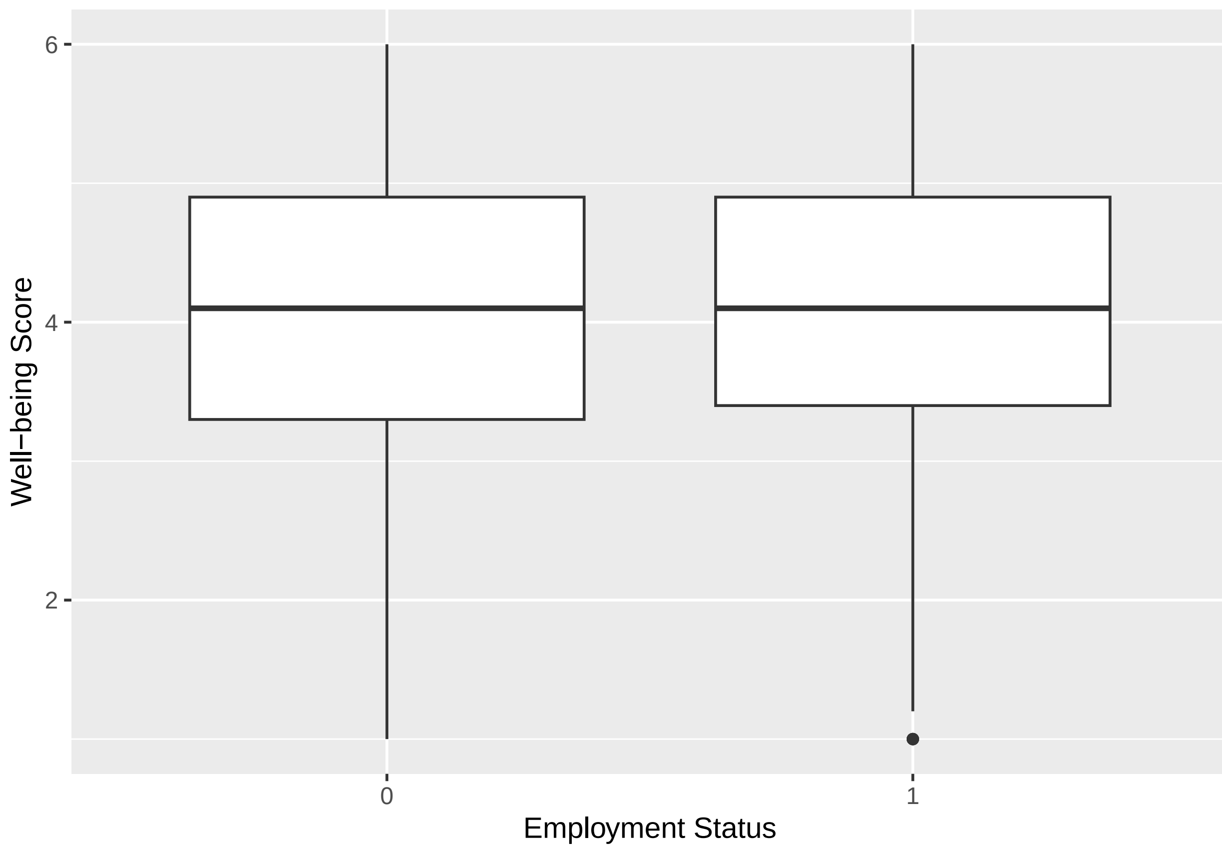

```
project_data %>%
  group_by(employed) %>%
  summarise(
    avg_wellbeing =
      mean(wellbeing_score,
           na.rm = TRUE)
  )
```

```
## # A tibble: 2 x 2
##   employed avg_wellbeing
##      <dbl>         <dbl>
## 1        0          4.03
## 2        1          4.09
```

```
#Employment vs Well-being
ggplot(
  project_data,
  aes(
    x = factor(employed),
    y = wellbeing_score
  )
) +
  geom_boxplot() +
  labs(
    x = "Employment Status",
```

```
    y = "Well-being Score"
  )
```

```
## Warning: Removed 4 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



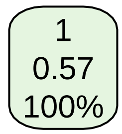

```
#Logistic Regression
#Does well-being predict employment?
wellbeing_logit <- glm(
  employed ~ wellbeing_score,
  data = project_data,
  family = binomial
)

summary(wellbeing_logit)
```

```
##
## Call:
## glm(formula = employed ~ wellbeing_score, family = binomial,
##     data = project_data)
##
## Coefficients:
##                    Estimate Std. Error z value Pr(>|z|)
## (Intercept)         0.09302    0.16184   0.575    0.565
```

```
## wellbeing_score  0.04898    0.03862   1.268    0.205
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 3317.5  on 2429  degrees of freedom
## Residual deviance: 3315.9  on 2428  degrees of freedom
##   (4 observations deleted due to missingness)
## AIC: 3319.9
##
## Number of Fisher Scoring iterations: 4
```

Higher well-being is associated with a slightly higher probability of being employed, but this relationship is weak and not statistically significant in the simple logit model.

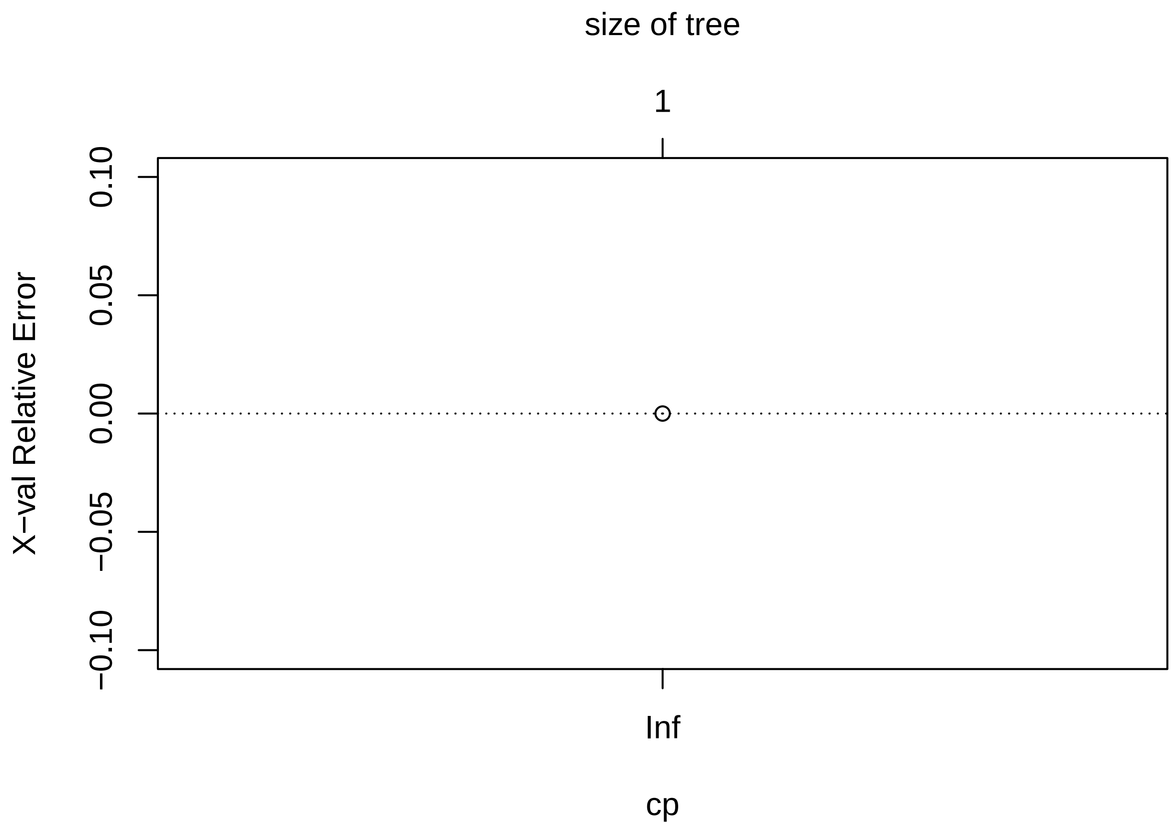
```r
#Tree
wellbeing_tree_data <- project_data %>%
  select(
    employed,
    all_of(wellbeing_vars)
  ) %>%
  na.omit()

colnames(wellbeing_tree_data) <- c(
  "employed",
  "Happy",
  "InterestInLife",
  "Satisfied",
  "Community",
  "SocietyBetter",
  "PeopleGood",
  "ManageResponsibilities",
  "TrustingRelationships",
  "Confidence",
  "LifePurpose"
)

wellbeing_tree <- rpart(
  employed ~ .,
  data = wellbeing_tree_data,
  method = "class"
)

rpart.plot(wellbeing_tree)
```




```r
printcp(wellbeing_tree)
```

```
##
## Classification tree:
## rpart(formula = employed ~ ., data = wellbeing_tree_data, method = "class")
##
## Variables actually used in tree construction:
## character(0)
##
## Root node error: 1025/2396 = 0.4278
##
## n= 2396
##
##          CP nsplit rel error xerror xstd
## 1 0.0078049      0         1      0    0
```

```r
plotcp(wellbeing_tree)
```

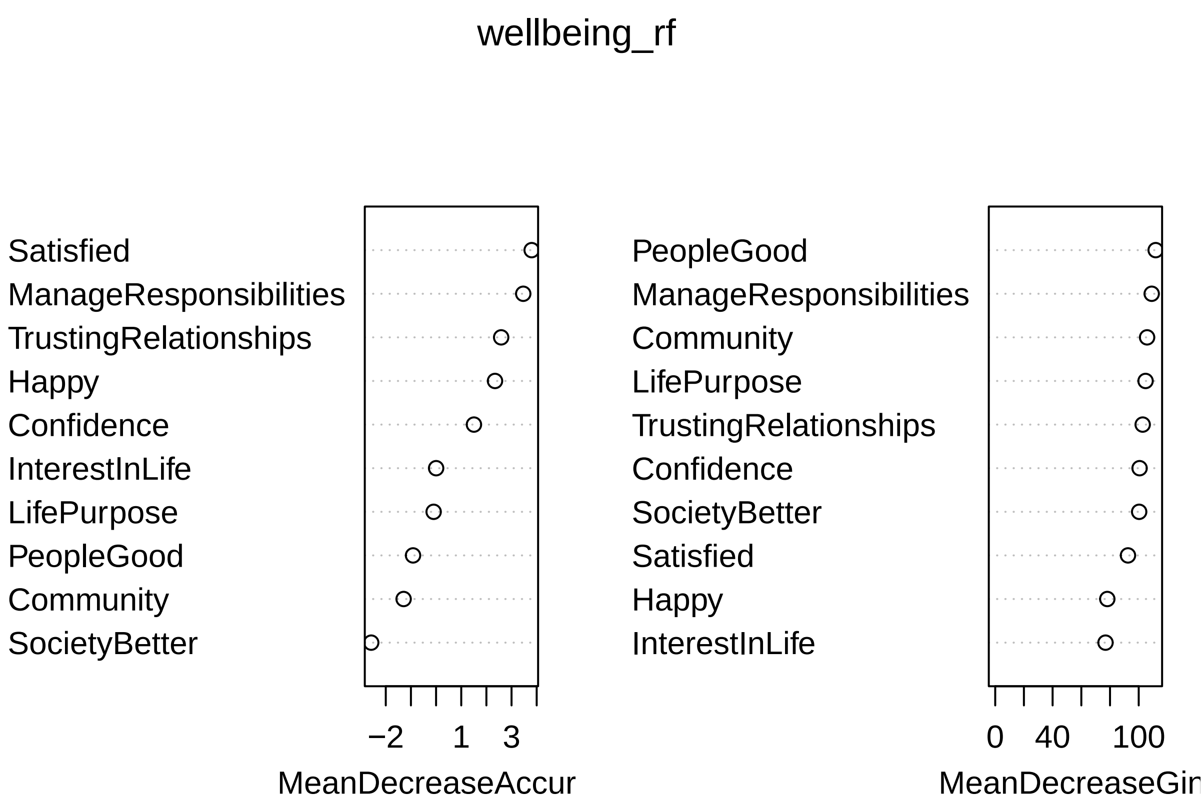


The CART model failed to identify meaningful splits, suggesting that individual well-being indicators alone do not strongly distinguish employed from non-employed young adults.

```r
#Random Forest
set.seed(123)

wellbeing_rf <- randomForest(
  as.factor(employed) ~ .,
  data = wellbeing_tree_data,
  ntree = 500,
  importance = TRUE
)

wellbeing_rf
```

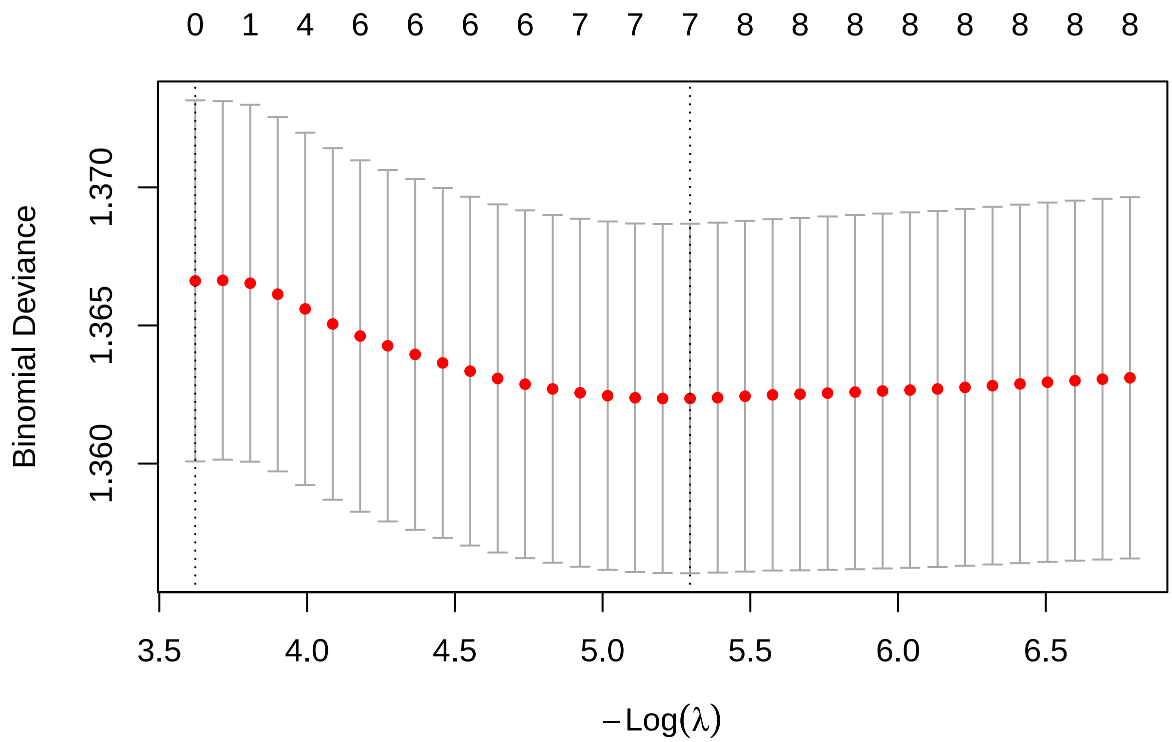
```
##
## Call:
##  randomForest(formula = as.factor(employed) ~ ., data = wellbeing_tree_data,      ntree = 500, impor
##                Type of random forest: classification
##                      Number of trees: 500
## No. of variables tried at each split: 3
##
##         OOB estimate of  error rate: 48.16%
## Confusion matrix:
##     0   1 class.error
## 0 302 723   0.7053659
## 1 431 940   0.3143691
```

```r
varImpPlot(wellbeing_rf)
```



Some well-being indicators appear moderately important, but predictive power remains weak.

```r
#LASSO
x_wellbeing <- model.matrix(
  employed ~ .,
  wellbeing_tree_data
)[, -1]

y_wellbeing <- wellbeing_tree_data$employed

set.seed(123)

wellbeing_lasso <- cv.glmnet(
  x_wellbeing,
  y_wellbeing,
  family = "binomial",
  alpha = 1
)

plot(wellbeing_lasso)
```




```r
coef(wellbeing_lasso, s = "lambda.min")
```

```
## 11 x 1 sparse Matrix of class "dgCMatrix"
##                           lambda.min
## (Intercept)             -0.102676366
## Happy                    0.030246278
## InterestInLife           0.032700109
## Satisfied                0.060432021
## Community               -0.032519379
## SocietyBetter           -0.069577728
## PeopleGood               .
## ManageResponsibilities   0.024485600
## TrustingRelationships    0.006561671
## Confidence               .
## LifePurpose              .
```

```r
coef(wellbeing_lasso, s = "lambda.1se")
```

```
## 11 x 1 sparse Matrix of class "dgCMatrix"
##                         lambda.1se
## (Intercept)              0.2908478
## Happy                    .
## InterestInLife           .
## Satisfied                .
```

```
## Community                  .
## SocietyBetter              .
## PeopleGood                 .
## ManageResponsibilities     .
## TrustingRelationships      .
## Confidence                 .
## LifePurpose                .
```

lambda.1se selected no predictors

Education

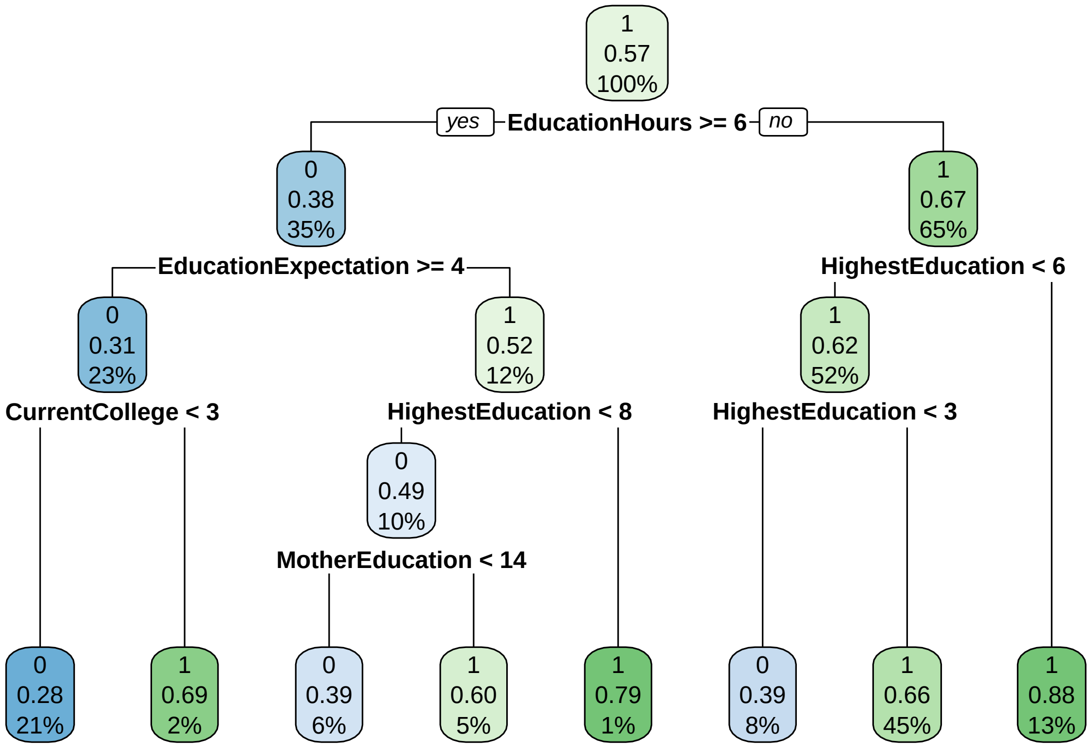
```r
education_vars <- c(
  "TA232393",
  "TA230990",
  "TA231000",
  "TA231001",
  "TA230986",
  "TA230988",
  "TA230685",
  "TA232394",
  "TA232396"
)

education_labels <- c(
  "HighestEducation",
  "GradeCompleted",
  "AttendedCollege",
  "CurrentCollege",
  "EducationAspiration",
  "EducationExpectation",
  "EducationHours",
  "MotherEducation",
  "FatherEducation"
)

education_tree_data <- project_data %>%
  select(
    employed,
    all_of(education_vars)
  ) %>%
  na.omit()

colnames(education_tree_data) <- c(
  "employed",
  education_labels
)
```

```r
# Logistic Regression
education_logit <- glm(
  employed ~ .,
  data = education_tree_data,
```

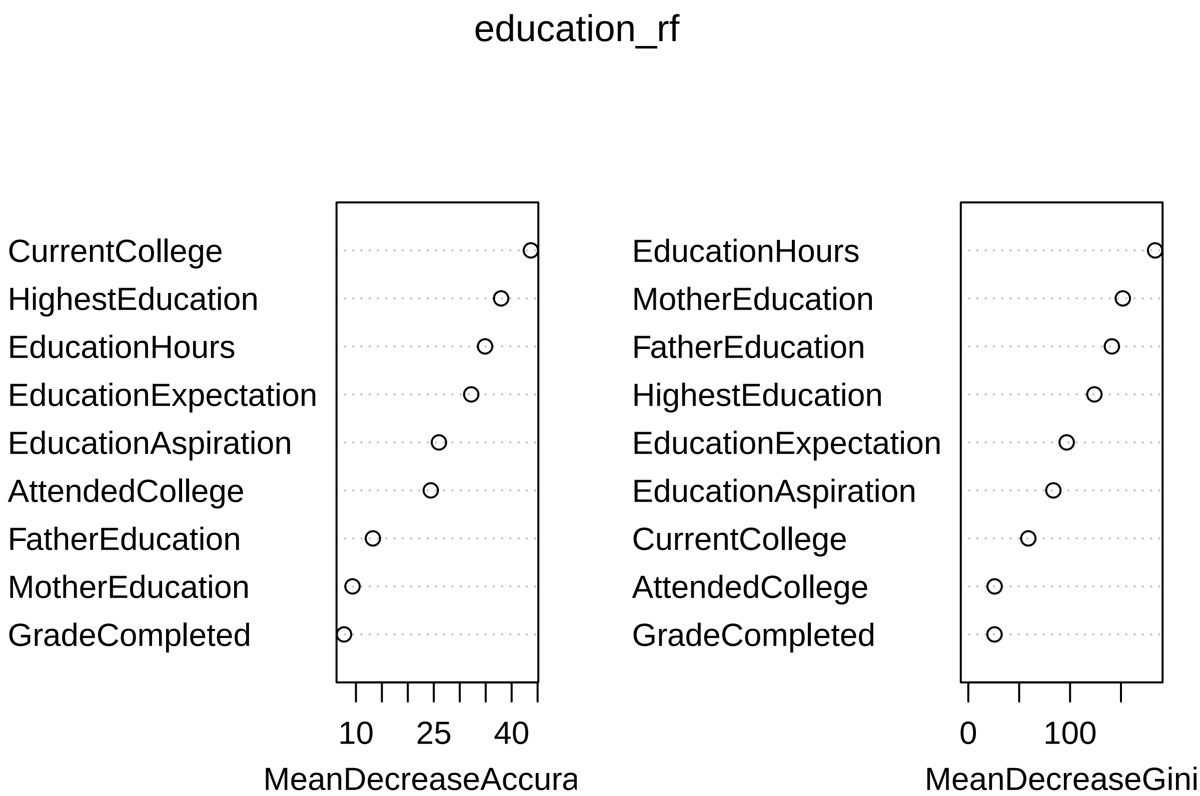
```r
  family = binomial
)

summary(education_logit)
```

```
##
## Call:
## glm(formula = employed ~ ., family = binomial, data = education_tree_data)
##
## Coefficients:
##                         Estimate Std. Error z value Pr(>|z|)
## (Intercept)           -0.9846252  0.1816047  -5.422  5.9e-08 ***
## HighestEducation       0.0108578  0.0046290   2.346 0.018998 *
## GradeCompleted         0.0112573  0.0108677   1.036 0.300272
## AttendedCollege        0.3564761  0.0342495  10.408  < 2e-16 ***
## CurrentCollege         0.4818661  0.0386155  12.479  < 2e-16 ***
## EducationAspiration  -0.0096882  0.0036026  -2.689 0.007162 **
## EducationExpectation -0.0728793  0.0204476  -3.564 0.000365 ***
## EducationHours        -0.0006538  0.0001731  -3.777 0.000159 ***
## MotherEducation       -0.0034145  0.0028676  -1.191 0.233763
## FatherEducation       -0.0035278  0.0012059  -2.925 0.003440 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 3324.3  on 2433  degrees of freedom
## Residual deviance: 3046.3  on 2424  degrees of freedom
## AIC: 3066.3
##
## Number of Fisher Scoring iterations: 4
```

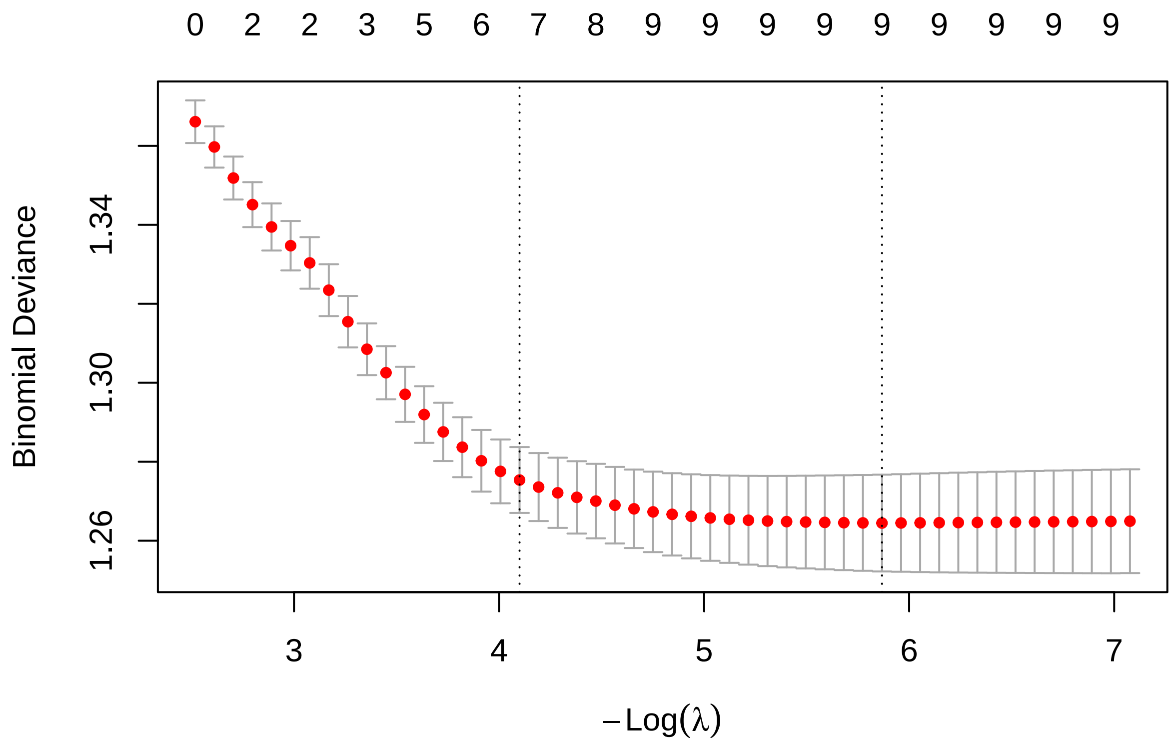
```r
#Tree
education_tree <- rpart(
  employed ~ .,
  data = education_tree_data,
  method = "class"
)

rpart.plot(education_tree)
```




```r
#Random Forest
set.seed(123)

education_rf <- randomForest(
  as.factor(employed) ~ .,
  data = education_tree_data,
  ntree = 500,
  importance = TRUE
)

varImpPlot(education_rf)
```




```r
x_education <- model.matrix(
  employed ~ .,
  education_tree_data
)[,-1]

y_education <- education_tree_data$employed

education_lasso <- cv.glmnet(
  x_education,
  y_education,
  family = "binomial",
  alpha = 1
)

plot(education_lasso)
```




```r
coef(
  education_lasso,
  s = "lambda.min"
)
```

```
## 10 x 1 sparse Matrix of class "dgCMatrix"
##                         lambda.min
## (Intercept)          -0.8842140325
## HighestEducation      0.0095812879
## GradeCompleted        0.0076598464
## AttendedCollege       0.3262841722
## CurrentCollege        0.4498574874
## EducationAspiration  -0.0079267475
## EducationExpectation -0.0739126671
## EducationHours       -0.0005955542
## MotherEducation      -0.0025398911
## FatherEducation      -0.0030499747
```

```r
coef(
  education_lasso,
  s = "lambda.1se"
)
```

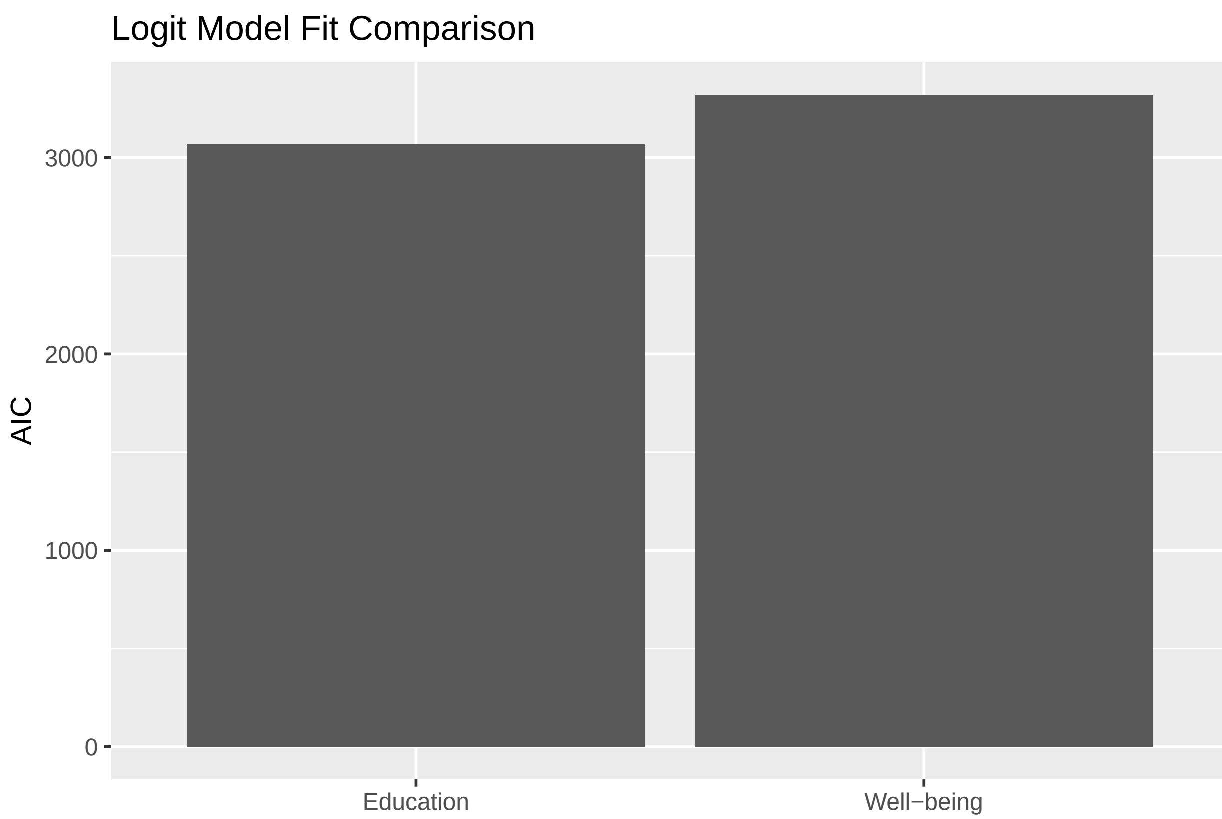
```
## 10 x 1 sparse Matrix of class "dgCMatrix"
##                         lambda.1se
```

```
## (Intercept)          -0.4212732199
## HighestEducation      0.0038430503
## GradeCompleted        .
## AttendedCollege       0.1926939205
## CurrentCollege        0.3080455834
## EducationAspiration  -0.0001424525
## EducationExpectation -0.0794257324
## EducationHours       -0.0003334004
## MotherEducation       .
## FatherEducation      -0.0009165377
```

```r
# Final comparison table

wellbeing_AIC <- AIC(wellbeing_logit)
education_AIC <- AIC(education_logit)

wellbeing_tree_splits <- wellbeing_tree$cptable[nrow(wellbeing_tree$cptable), "nsplit"]
education_tree_splits <- education_tree$cptable[nrow(education_tree$cptable), "nsplit"]

wellbeing_lasso_1se <- coef(wellbeing_lasso, s = "lambda.1se")
education_lasso_1se <- coef(education_lasso, s = "lambda.1se")

wellbeing_lasso_vars <- sum(wellbeing_lasso_1se[-1, ] != 0)
education_lasso_vars <- sum(education_lasso_1se[-1, ] != 0)

wellbeing_rf_acc <- mean(
  predict(wellbeing_rf, wellbeing_tree_data) ==
    wellbeing_tree_data$employed
)

education_rf_acc <- mean(
  predict(education_rf, education_tree_data) ==
    education_tree_data$employed
)

comparison_table <- data.frame(
  Model_Group = c("Well-being", "Education"),
  Logit_AIC = c(wellbeing_AIC, education_AIC),
  Tree_Splits = c(wellbeing_tree_splits, education_tree_splits),
  LASSO_lambda_1se_Selected_Variables =
    c(wellbeing_lasso_vars, education_lasso_vars),
  Random_Forest_Training_Accuracy =
    c(wellbeing_rf_acc, education_rf_acc)
)

comparison_table
```

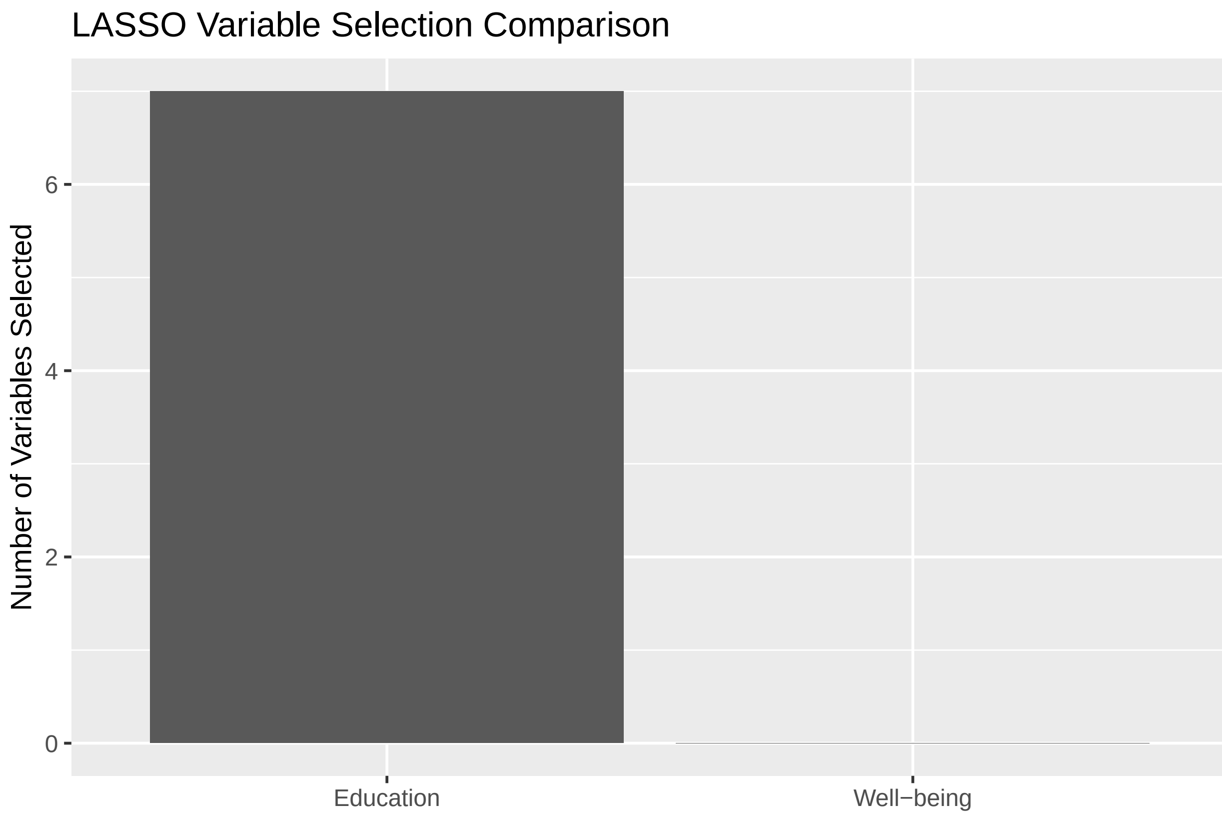
```
##   Model_Group Logit_AIC Tree_Splits LASSO_lambda_1se_Selected_Variables
## 1  Well-being  3319.918           0                                   0
## 2   Education  3066.293           7                                   7
##   Random_Forest_Training_Accuracy
## 1                       0.9545075
## 2                       0.9182416
```

```r
ggplot(
  comparison_table,
  aes(
    x = Model_Group,
    y = Logit_AIC
  )
) +
  geom_col() +
  labs(
    title = "Logit Model Fit Comparison",
    x = "",
    y = "AIC"
  )
```




```r
ggplot(
  comparison_table,
  aes(
    x = Model_Group,
    y = LASSO_lambda_1se_Selected_Variables
  )
) +
  geom_col() +
  labs(
    title = "LASSO Variable Selection Comparison",
    x = "",
```

```
    y = "Number of Variables Selected"
  )
```



Overall, the education variables provide stronger and more stable predictive information than the well-being variables. The education logit model has a much lower AIC, the education tree identifies meaningful splits, and LASSO retains several education variables under the conservative lambda.1se rule. In contrast, the well-being tree does not split and LASSO removes all well-being variables at lambda.1se. Although the random forest can fit both sets of variables, the broader pattern suggests that education is the stronger predictor of employment outcomes.